

Quiz 1 - Instructor key (revision 1)

Five one-page forms; 10 minutes; 3 points. One easy expected-value item, one medium mean/variance item, and one hard iterated-expectations item per form. Every item has different numerical inputs and a different correct value on all five forms.

Form	Q1	Q2	Q3	Item order
B7CS	C	D	F	Easy / Medium / Hard
UF5C	E	B	D	Medium / Hard / Easy
P76Y	E	G	A	Hard / Easy / Medium
THZ2	F	A	C	Easy / Hard / Medium
PEY5	G	B	D	Medium / Easy / Hard

Form B7CS - worked solutions

Question 1 (easy) - C: 4

A random variable X equals 2 with probability 0.75 and equals 10 with probability 0.25. What is $E(X)$?

$$E(X) = 0.75 \times 2 + 0.25 \times 10 = 4.$$

Question 2 (medium) - D: (10, 18)

Suppose $E(Y) = 2$, $\text{Var}(Y) = 2$, and $X = 3Y + 4$. Which ordered pair gives $(E(X), \text{Var}(X))$?

$$E(X) = 3 \times 2 + 4 = 10. \text{Var}(X) = 3^2 \times 2 = 18; \text{ adding 4 does not change the variance.}$$

Question 3 (hard) - F: 26

Let D be 0 or 1 and W be hourly earnings in euros. In population A, $\Pr(D = 1) = 0.25$, $E(W) = 18$, and $E(W \mid D = 0) = 14$. In population B, $\Pr(D = 1) = 0.75$. The conditional means $E(W \mid D = 0)$ and $E(W \mid D = 1)$ are the same in both populations. What is $E(W)$ in population B, in euros per hour?

Write $m = E(W \mid D = 1)$. Iterated expectations in A give $18 = 0.25m + 0.75 \times 14$, so $m = 30$. Reweight the same conditional means in B: $E(W) = 0.75 \times 30 + 0.25 \times 14 = 26$ euros per hour.

Form UF5C - worked solutions

Question 1 (medium) - E: (11, 12)

Suppose $E(Y) = 3$, $\text{Var}(Y) = 3$, and $X = 2Y + 5$. Which ordered pair gives $(E(X), \text{Var}(X))$?

$E(X) = 2 \times 3 + 5 = 11$. $\text{Var}(X) = 2^2 \times 3 = 12$; adding 5 does not change the variance.

Question 2 (hard) - B: 16

Let D be 0 or 1 and W be hourly earnings in euros. In population A, $\Pr(D = 1) = 0.75$, $E(W) = 24$, and $E(W | D = 0) = 12$. In population B, $\Pr(D = 1) = 0.25$. The conditional means $E(W | D = 0)$ and $E(W | D = 1)$ are the same in both populations. What is $E(W)$ in population B, in euros per hour?

Write $m = E(W | D = 1)$. Iterated expectations in A give $24 = 0.75m + 0.25 \times 12$, so $m = 28$. Reweight the same conditional means in B: $E(W) = 0.25 \times 28 + 0.75 \times 12 = 16$ euros per hour.

Question 3 (easy) - D: 6

A random variable X equals 3 with probability 0.75 and equals 15 with probability 0.25. What is $E(X)$?

$E(X) = 0.75 \times 3 + 0.25 \times 15 = 6$.

Form P76Y - worked solutions

Question 1 (hard) - E: 34

Let D be 0 or 1 and W be hourly earnings in euros. In population A, $\Pr(D = 1) = 0.25$, $E(W) = 22$, and $E(W | D = 0) = 16$. In population B, $\Pr(D = 1) = 0.75$. The conditional means $E(W | D = 0)$ and $E(W | D = 1)$ are the same in both populations. What is $E(W)$ in population B, in euros per hour?

Write $m = E(W | D = 1)$. Iterated expectations in A give $22 = 0.25m + 0.75 \times 16$, so $m = 40$. Reweight the same conditional means in B: $E(W) = 0.75 \times 40 + 0.25 \times 16 = 34$ euros per hour.

Question 2 (easy) - G: 10

A random variable X equals 4 with probability 0.25 and equals 12 with probability 0.75. What is $E(X)$?

$$E(X) = 0.25 \times 4 + 0.75 \times 12 = 10.$$

Question 3 (medium) - A: (14, 36)

Suppose $E(Y) = 4$, $\text{Var}(Y) = 4$, and $X = 3Y + 2$. Which ordered pair gives $(E(X), \text{Var}(X))$?

$$E(X) = 3 \times 4 + 2 = 14. \text{Var}(X) = 3^2 \times 4 = 36; \text{ adding 2 does not change the variance.}$$

Form THZ2 - worked solutions

Question 1 (easy) - F: 12

A random variable X equals 6 with probability 0.25 and equals 14 with probability 0.75. What is $E(X)$?

$$E(X) = 0.25 \times 6 + 0.75 \times 14 = 12.$$

Question 2 (hard) - A: 18

Let D be 0 or 1 and W be hourly earnings in euros. In population A, $\Pr(D = 1) = 0.75$, $E(W) = 34$, and $E(W | D = 0) = 10$. In population B, $\Pr(D = 1) = 0.25$. The conditional means $E(W | D = 0)$ and $E(W | D = 1)$ are the same in both populations. What is $E(W)$ in population B, in euros per hour?

Write $m = E(W | D = 1)$. Iterated expectations in A give $34 = 0.75m + 0.25 \times 10$, so $m = 42$. Reweight the same conditional means in B: $E(W) = 0.25 \times 42 + 0.75 \times 10 = 18$ euros per hour.

Question 3 (medium) - C: (13, 20)

Suppose $E(Y) = 5$, $\text{Var}(Y) = 5$, and $X = 2Y + 3$. Which ordered pair gives $(E(X), \text{Var}(X))$?

$$E(X) = 2 \times 5 + 3 = 13. \text{Var}(X) = 2^2 \times 5 = 20; \text{ adding 3 does not change the variance.}$$

Form PEY5 - worked solutions

Question 1 (medium) - G: (18, 48)

Suppose $E(Y) = 3$, $\text{Var}(Y) = 3$, and $X = 4Y + 6$. Which ordered pair gives $(E(X), \text{Var}(X))$?

$E(X) = 4 \times 3 + 6 = 18$. $\text{Var}(X) = 4^2 \times 3 = 48$; adding 6 does not change the variance.

Question 2 (easy) - B: 11

A random variable X equals 8 with probability 0.75 and equals 20 with probability 0.25. What is $E(X)$?

$E(X) = 0.75 \times 8 + 0.25 \times 20 = 11$.

Question 3 (hard) - D: 42

Let D be 0 or 1 and W be hourly earnings in euros. In population A, $\Pr(D = 1) = 0.25$, $E(W) = 26$, and $E(W \mid D = 0) = 18$. In population B, $\Pr(D = 1) = 0.75$. The conditional means $E(W \mid D = 0)$ and $E(W \mid D = 1)$ are the same in both populations. What is $E(W)$ in population B, in euros per hour?

Write $m = E(W \mid D = 1)$. Iterated expectations in A give $26 = 0.25m + 0.75 \times 18$, so $m = 50$. Reweight the same conditional means in B: $E(W) = 0.75 \times 50 + 0.25 \times 18 = 42$ euros per hour.